

Loxahatchee River-Lake Worth Creek Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 05 May, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

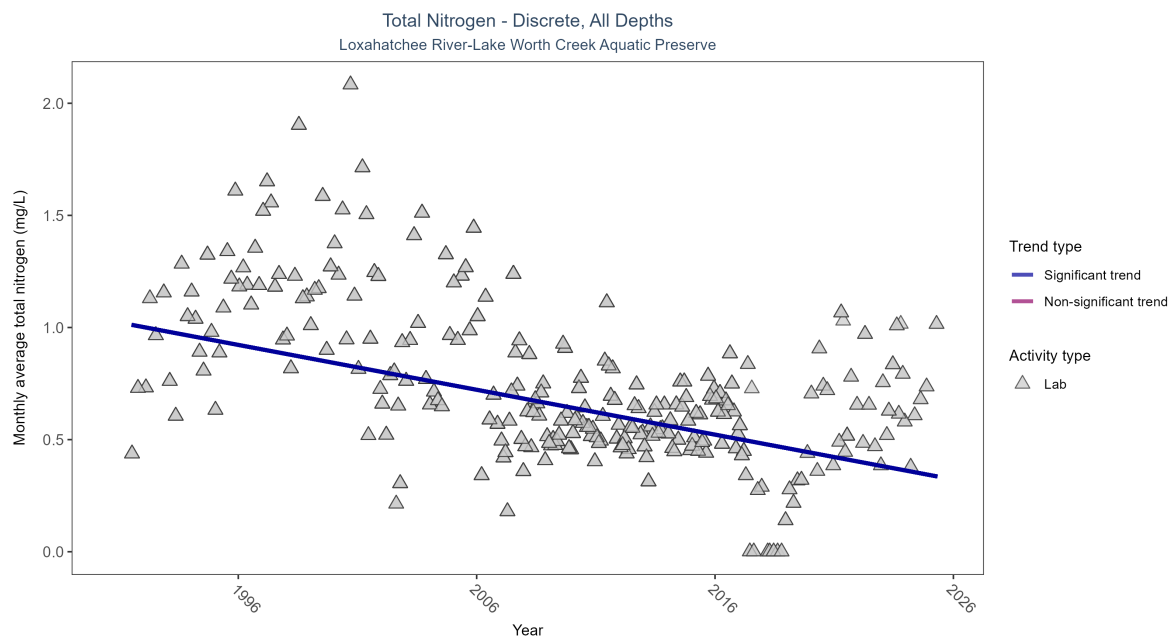


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Seasonal Kendall-Tau Results for - Total Nitrogen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly decreasing trend	2796	35	1991 - 2025	0.71	-0.31658	1.02239	-0.02001	0

Monthly average total nitrogen decreased by 0.02 mg/L per year.

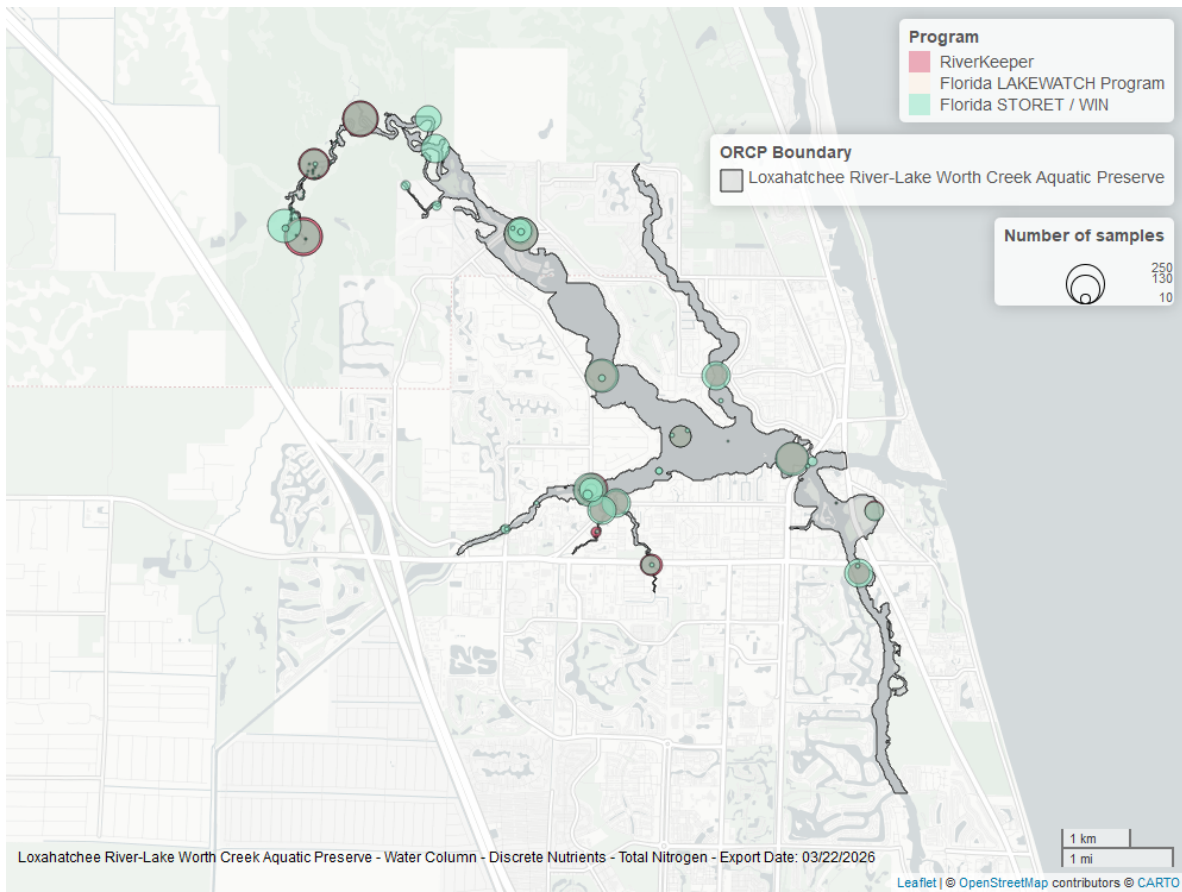


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

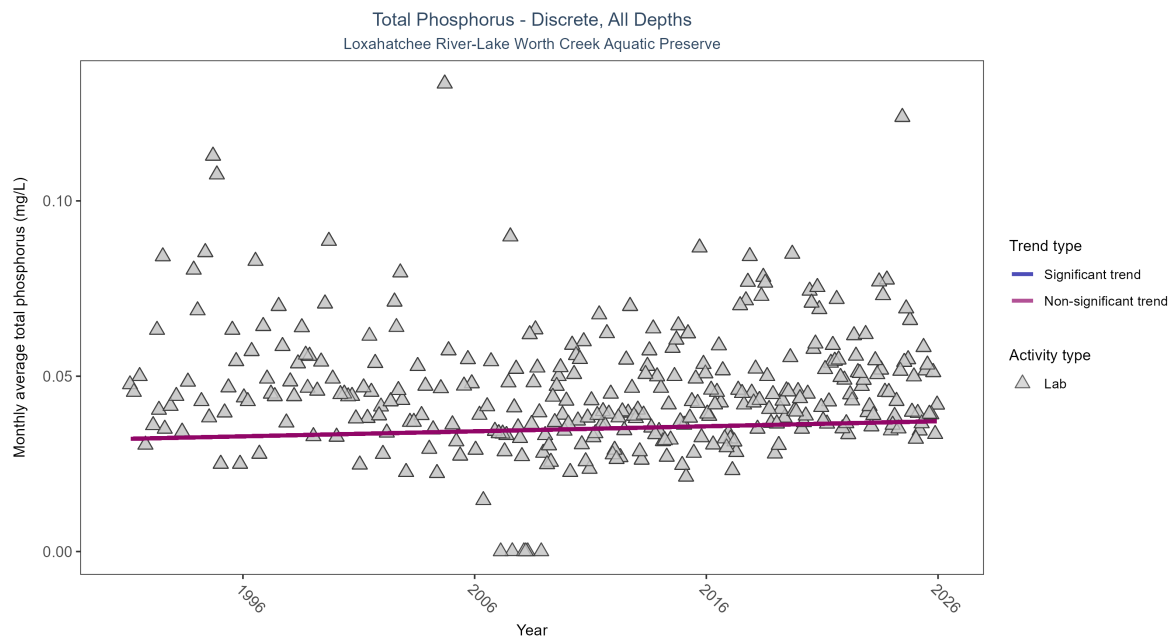


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Seasonal Kendall-Tau Results for - Total Phosphorus

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	4252	35	1991 - 2025	0.043	0.09959	0.03214	0.00014	0.072

Total phosphorus showed no detectable trend between 1991 and 2025.

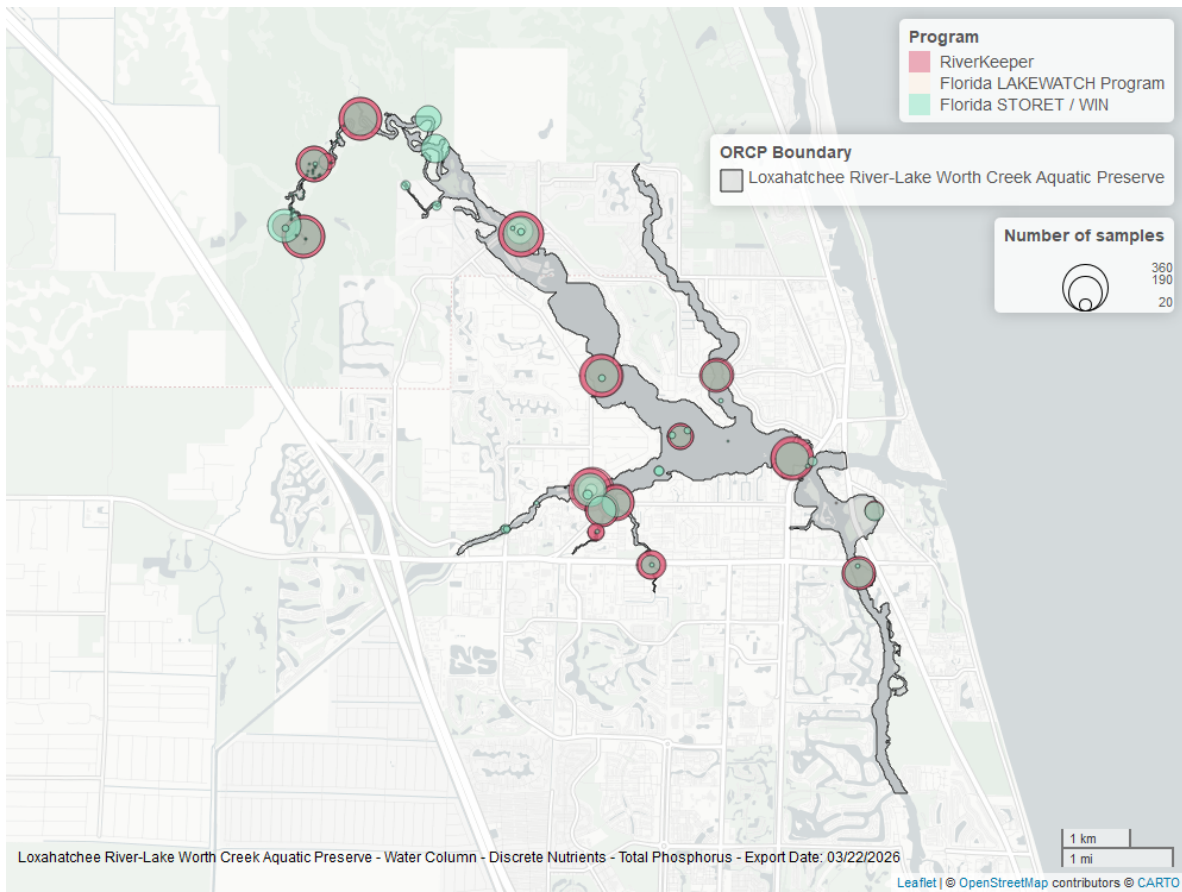


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Discrete

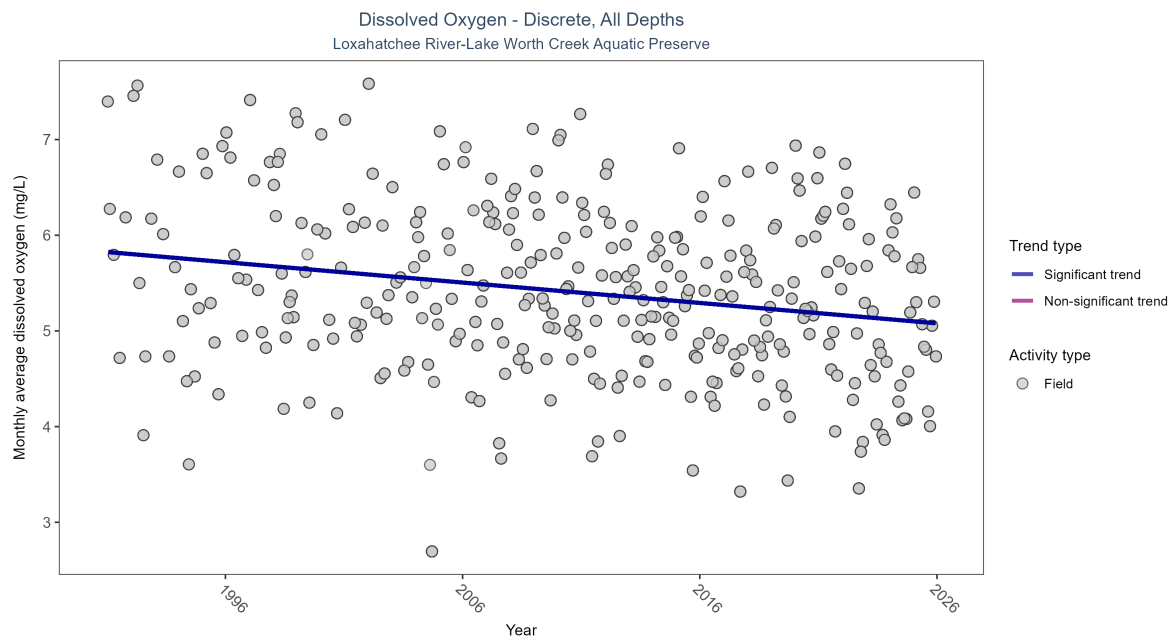


Figure 5: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 3: Seasonal Kendall-Tau Results for - Dissolved Oxygen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	8926	35	1991 - 2025	5.58	-0.25094	5.82491	-0.0213	0

Monthly average dissolved oxygen decreased by 0.02 mg/L per year.

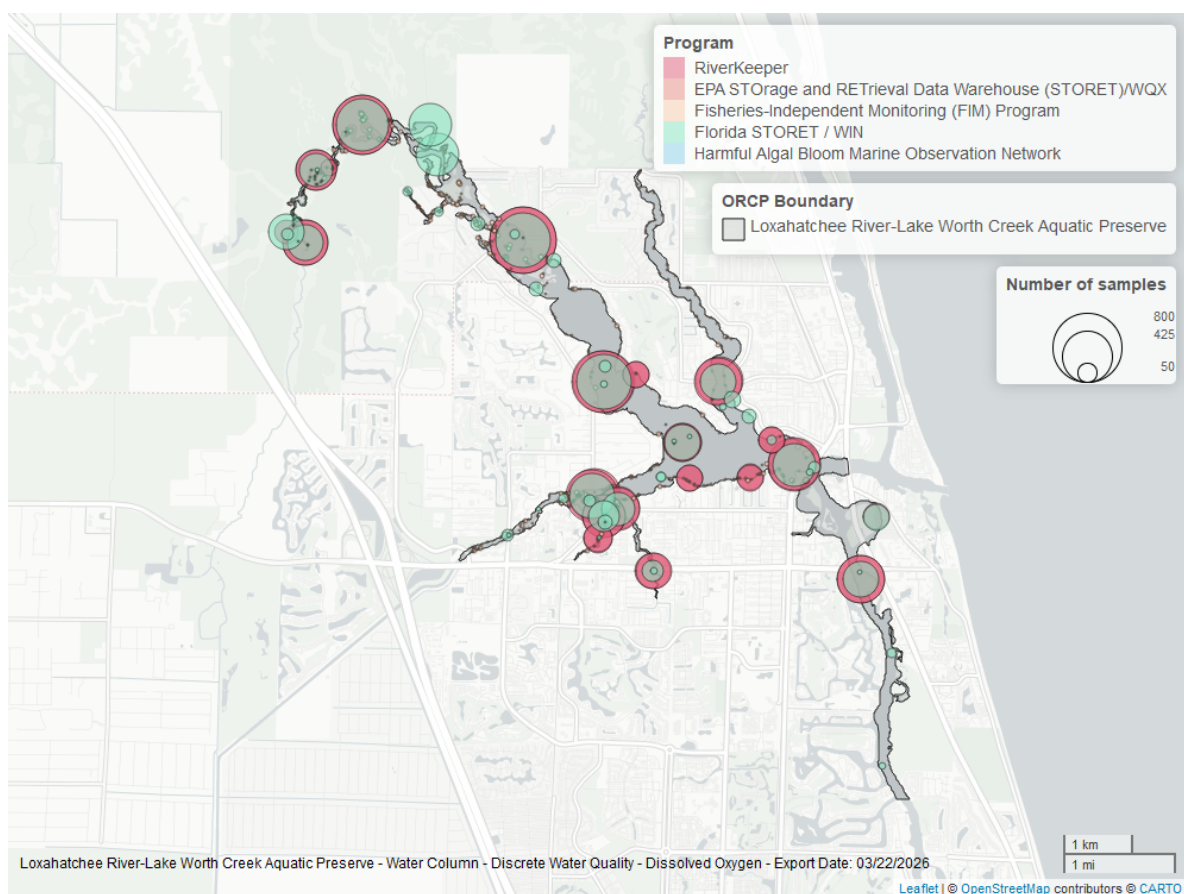


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

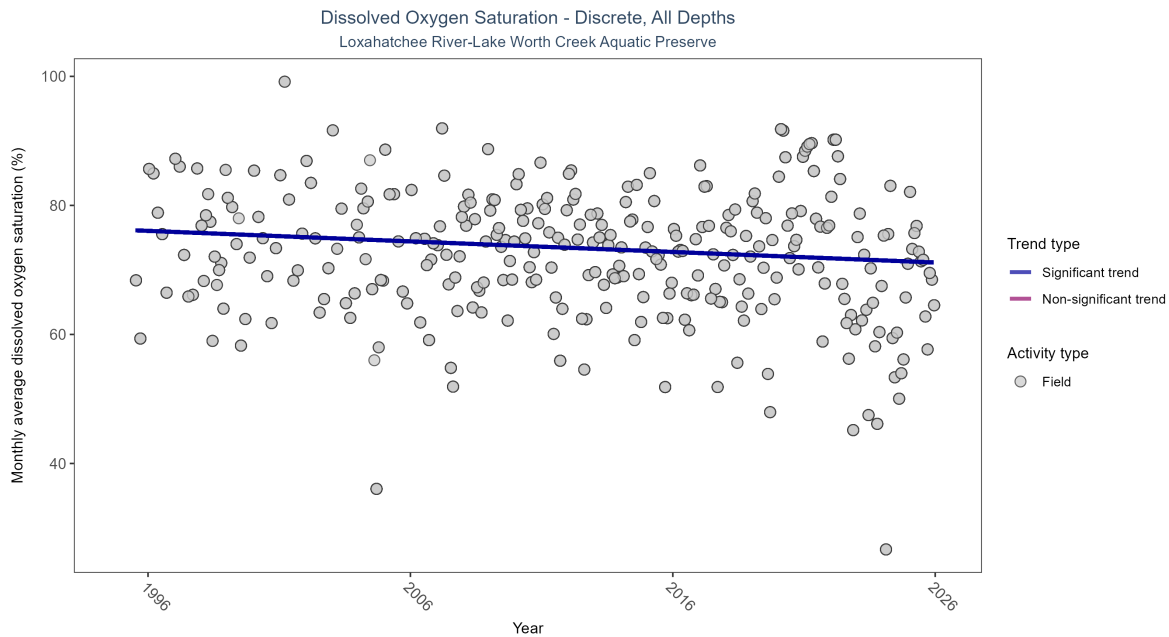


Figure 7: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 4: Seasonal Kendall-Tau Results for - Dissolved Oxygen Saturation

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	7231	31	1995 - 2025	75.2	-0.12904	76.21659	-0.16346	0.0013

Monthly average dissolved oxygen saturation decreased by 0.16% per year.

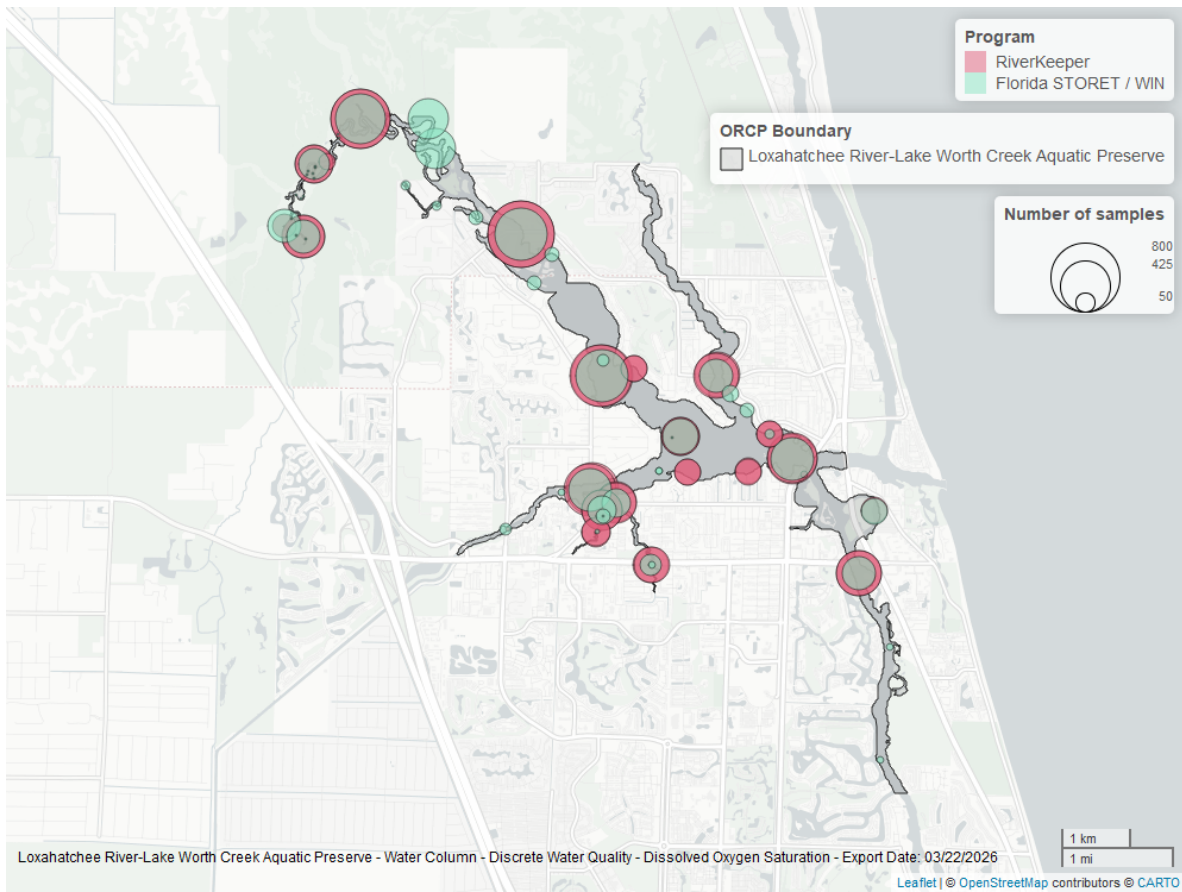


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

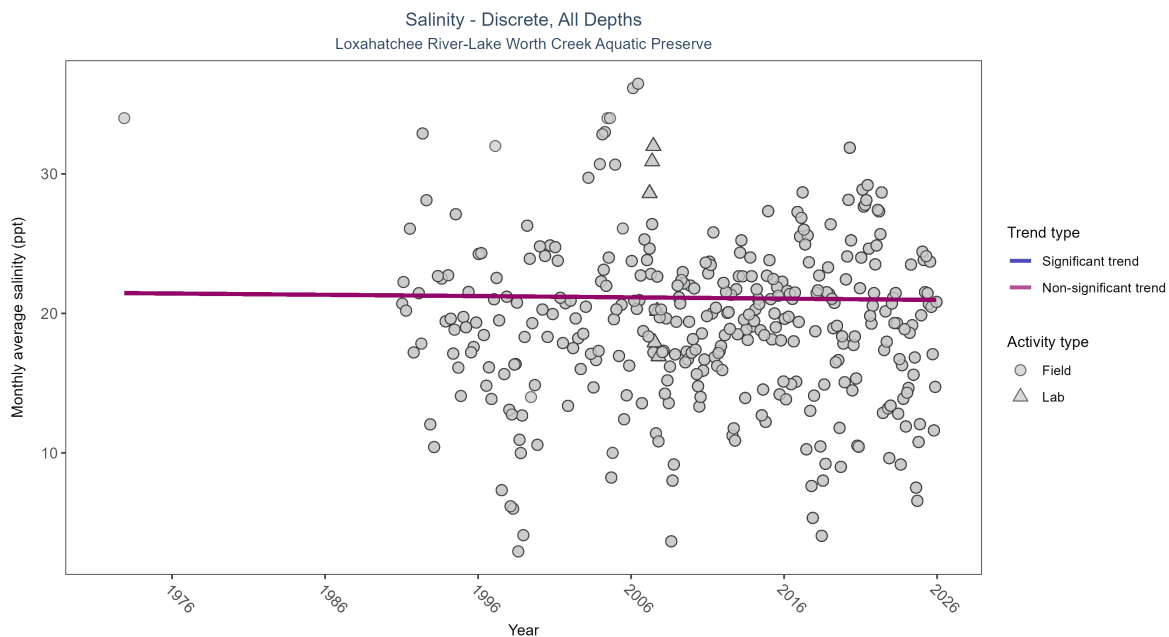


Figure 9: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 5: Seasonal Kendall-Tau Results for - Salinity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	No significant trend	12306	36	1972 - 2025	24.4	-0.00982	21.46318	-0.00921	0.7936

Salinity showed no detectable trend between 1972 and 2025.

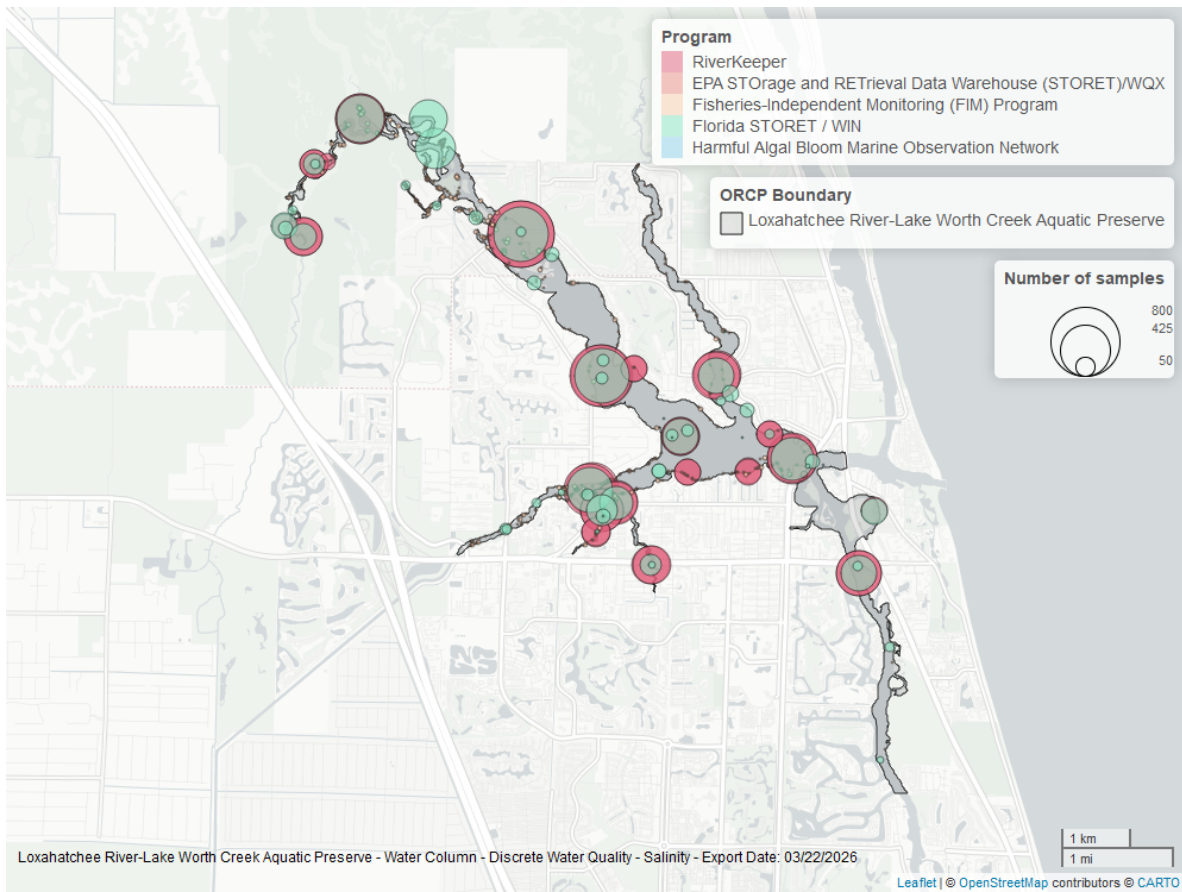


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

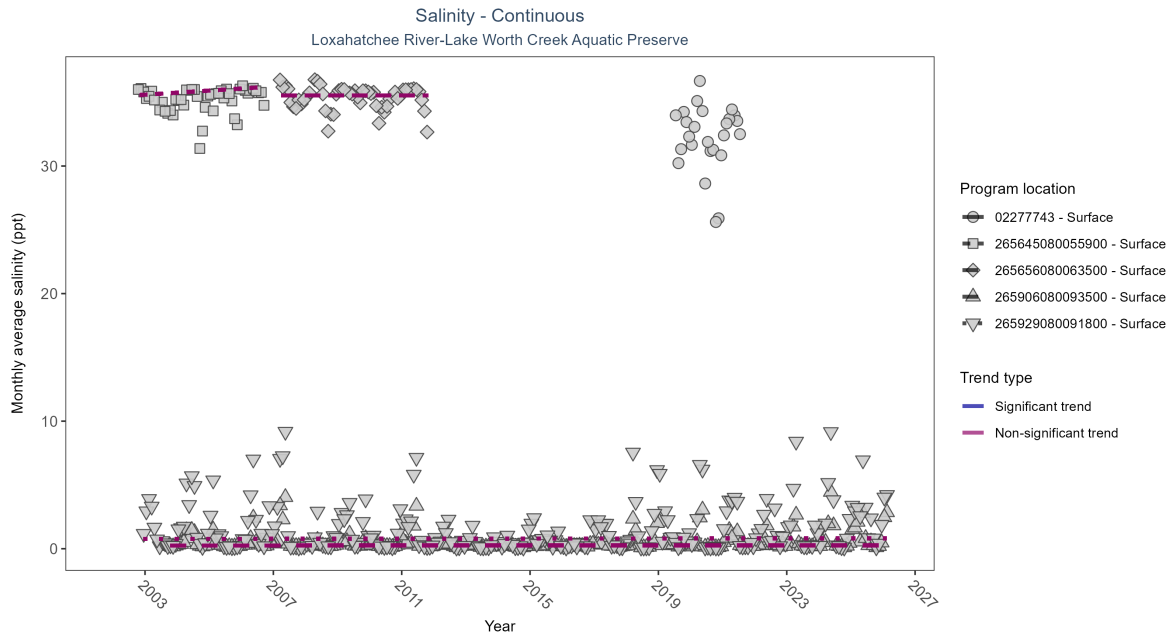


Figure 11: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 6: Seasonal Kendall-Tau Results - Salinity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02277743	Insufficient data to calculate trend	1144	3	2019 - 2021	32.0	-	-	-	-
265645080055900	No significant trend	1354	5	2002 - 2006	36.0	0.22	35.43	0.16	0.1594
265656080063500	No significant trend	2046	5	2007 - 2011	36.0	0.01	35.53	0	1
265906080093500	No significant trend	9559	24	2003 - 2026	0.3	0.05	0.25	0	0.264
265929080091800	No significant trend	8275	25	2002 - 2026	0.4	0.04	0.76	0	0.3874

No detectable change in monthly average salinity was observed at four locations. There was insufficient data to fit a model for one location.

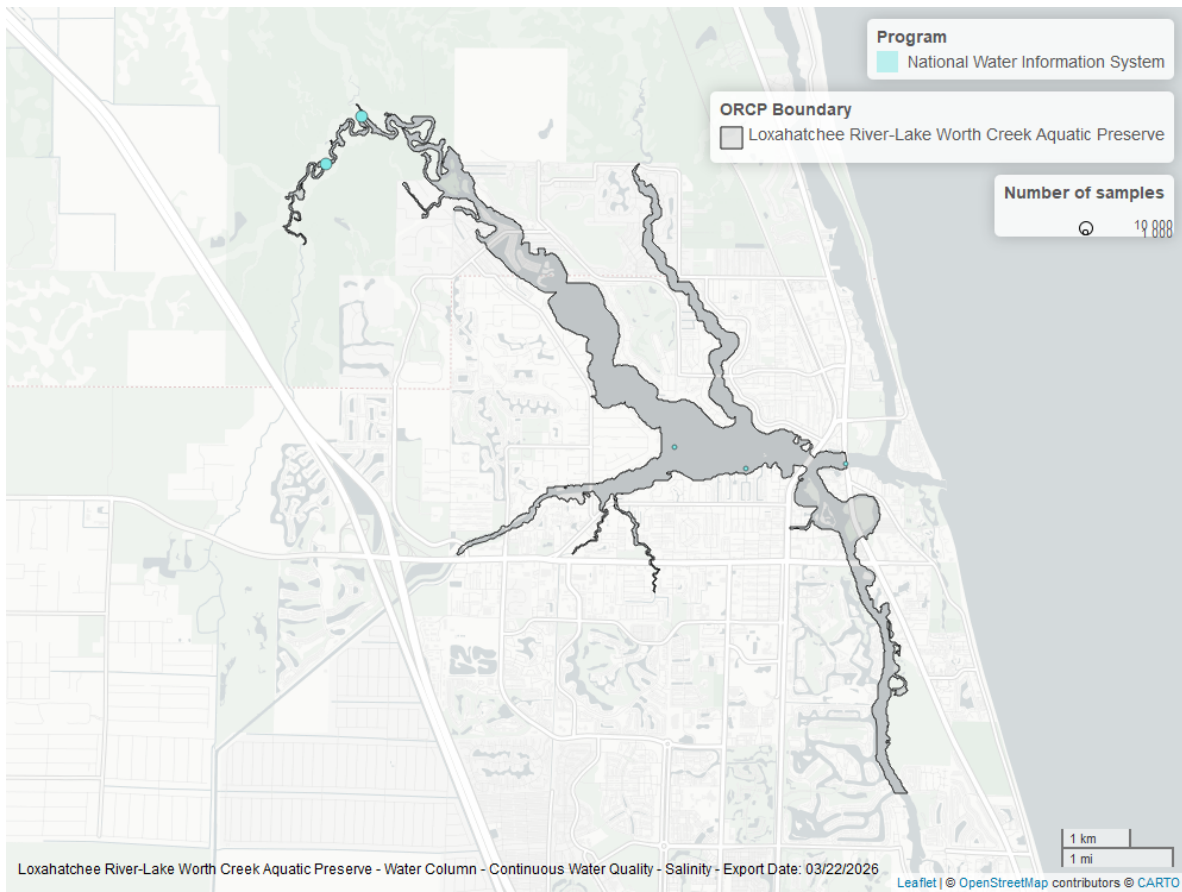


Figure 12: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

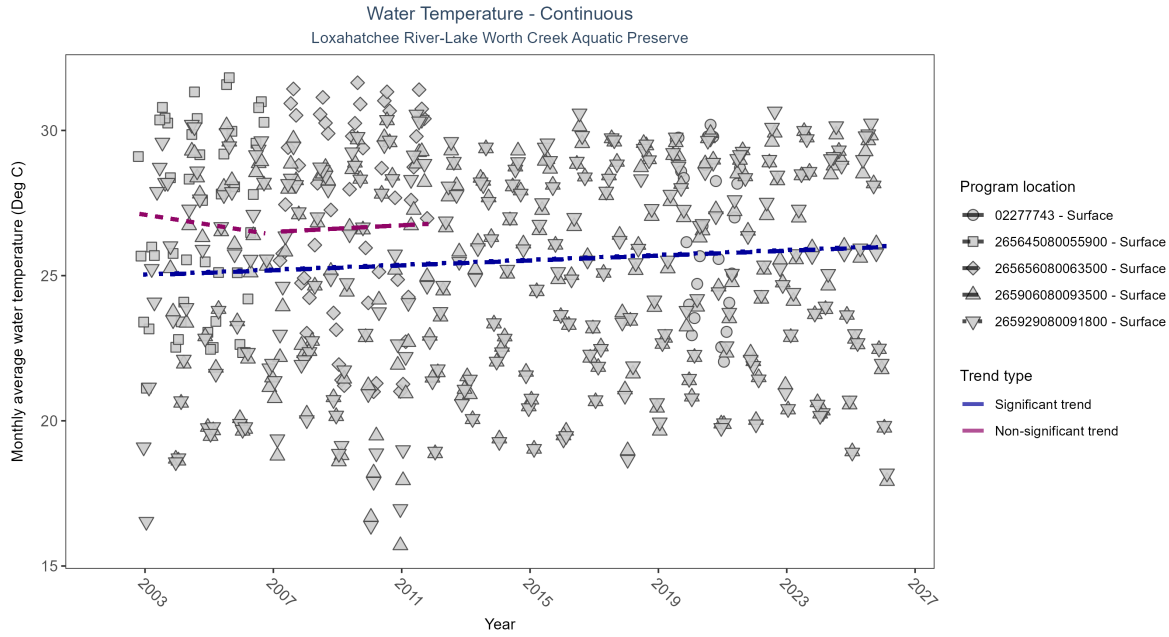


Figure 13: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 7: Seasonal Kendall-Tau Results - Water Temperature

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02277743	Insufficient data to calculate trend	1122	3	2019 - 2021	26.3	-	-	-	-
265645080055900	No significant trend	1354	5	2002 - 2006	26.6	-0.29	27.27	-0.17	0.0704
265656080063500	No significant trend	2664	5	2007 - 2011	27.6	0.08	26.5	0.06	0.4533
265906080093500	Significantly increasing trend	12641	24	2003 - 2026	25.6	0.27	25.01	0.04	0
265929080091800	Significantly increasing trend	8297	25	2002 - 2026	25.7	0.23	25	0.04	0

At two program locations, monthly average water temperature increased by 0.04°C per year. No detectable change in monthly average water temperature was observed at two locations. There was insufficient data to fit a model for one location.

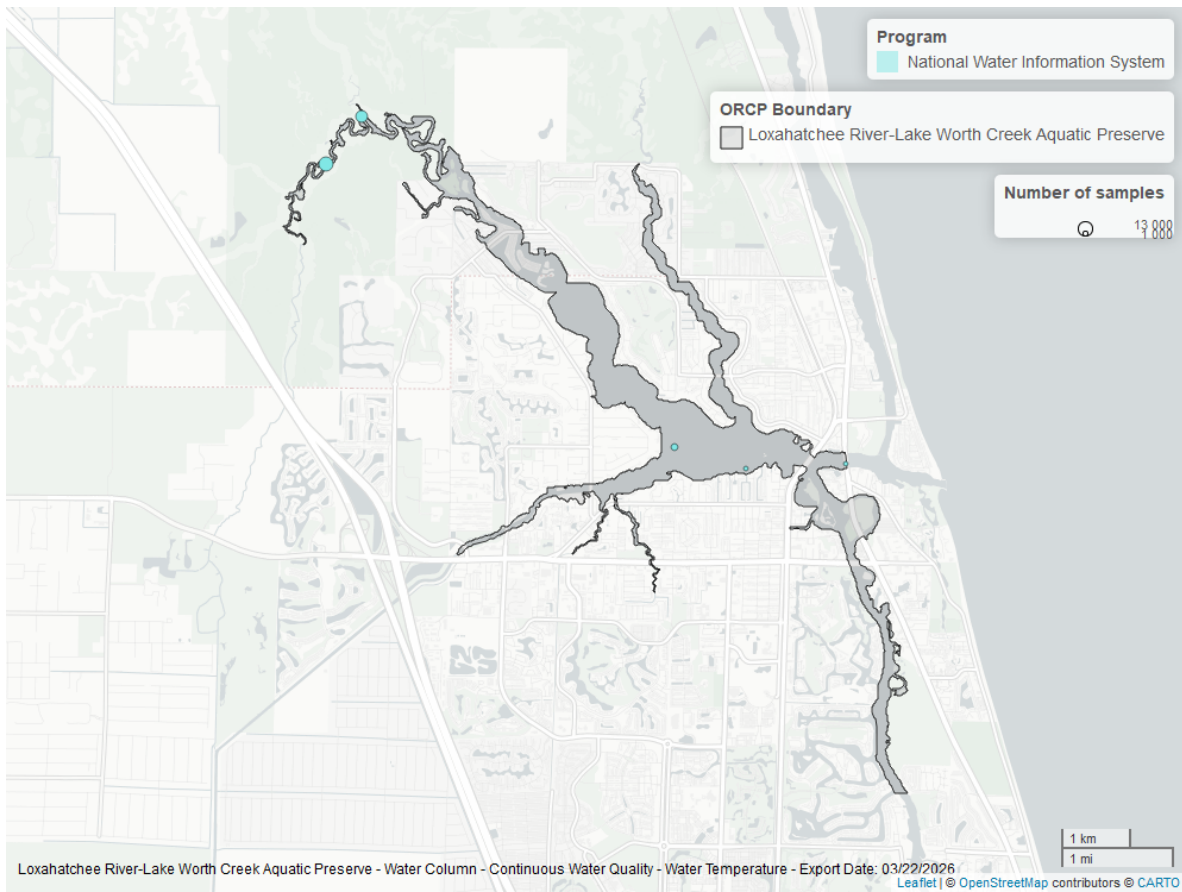


Figure 14: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

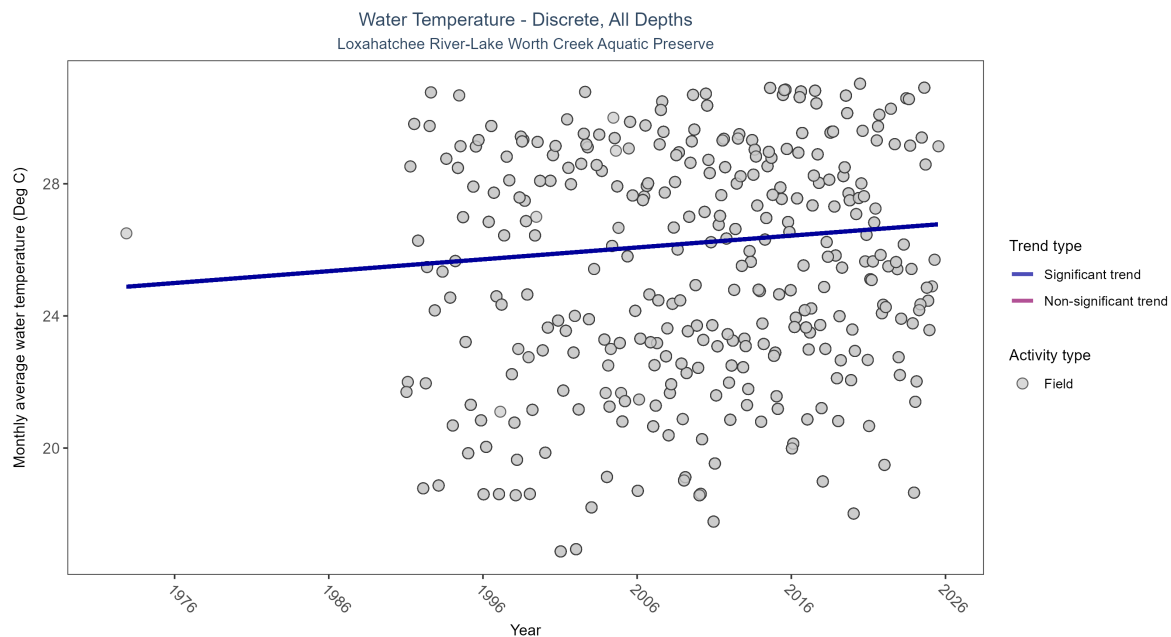


Figure 15: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 8: Seasonal Kendall-Tau Results for - Water Temperature

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly increasing trend	7069	36	1972 - 2025	25.5	0.17431	24.85234	0.03588	0

Monthly average water temperature increased by 0.04°C per year.

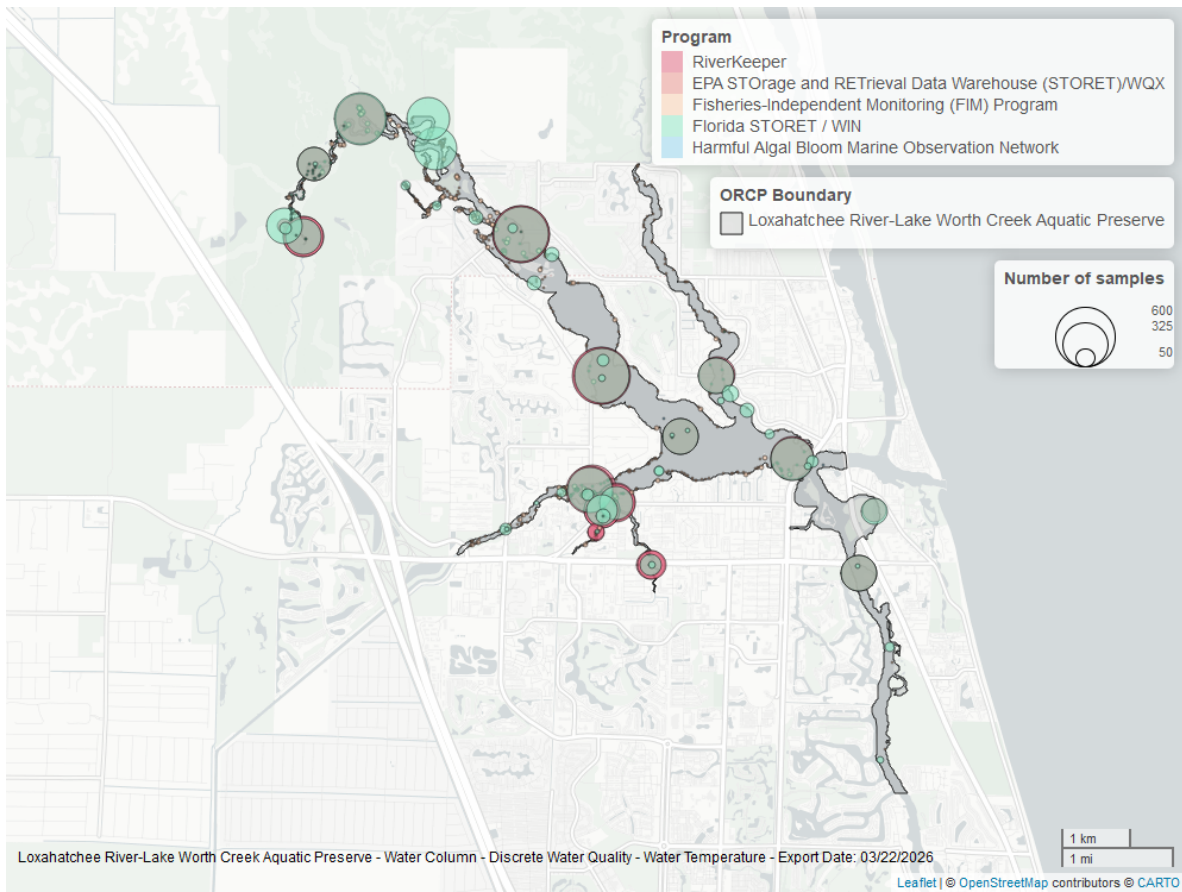


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

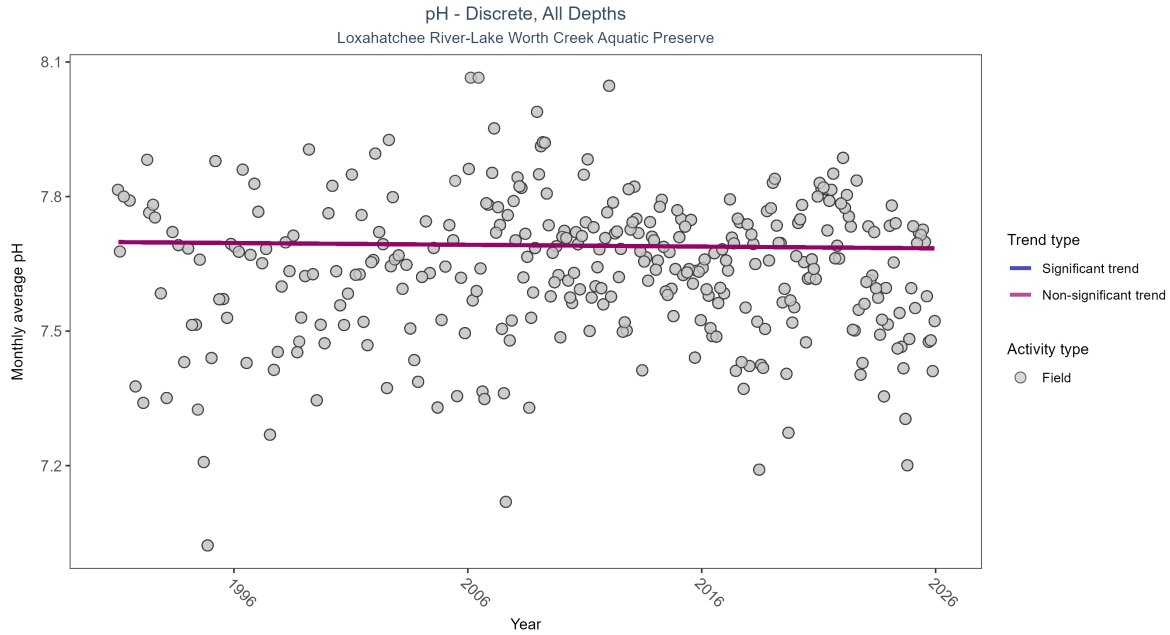


Figure 17: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 9: Seasonal Kendall-Tau Results for - pH

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	8835	35	1991 - 2025	7.69	-0.0411	7.69837	-0.0004	0.5682

pH showed no detectable trend between 1991 and 2025.

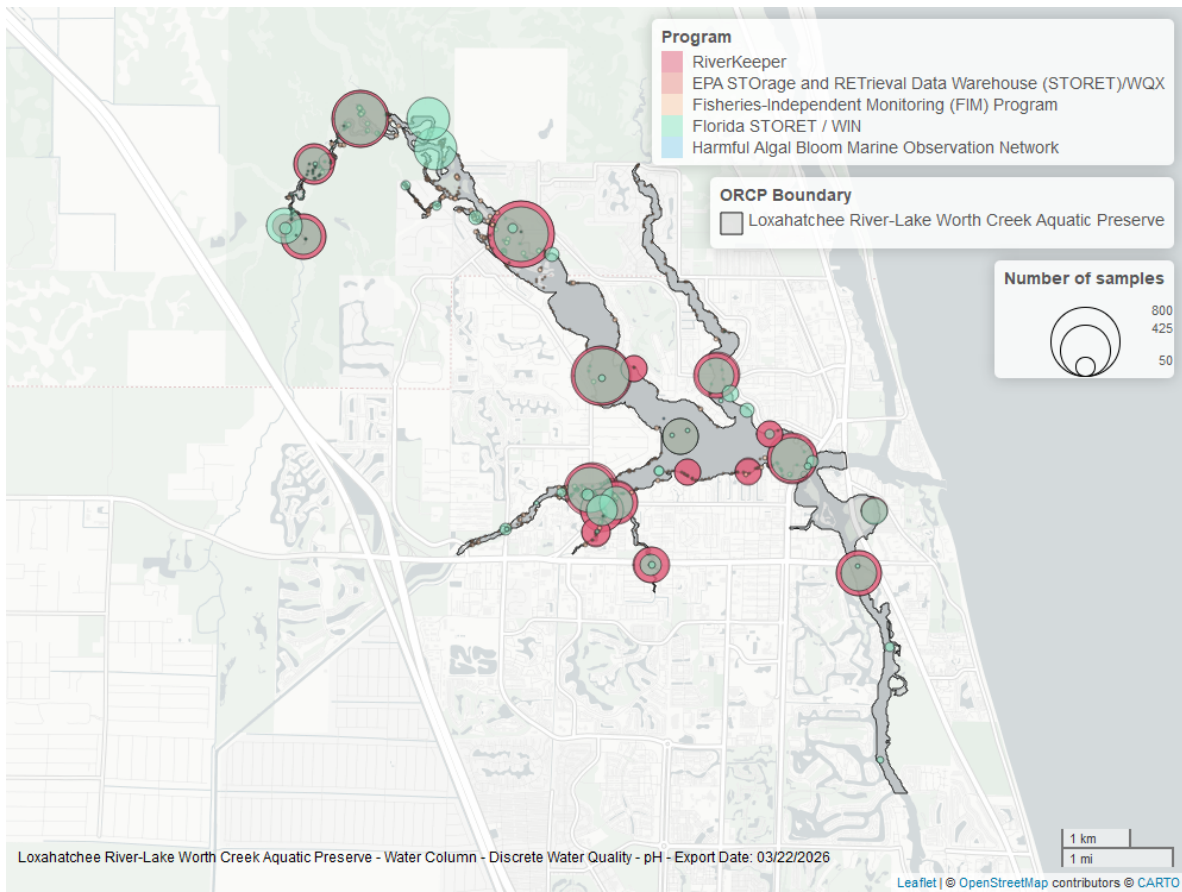


Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

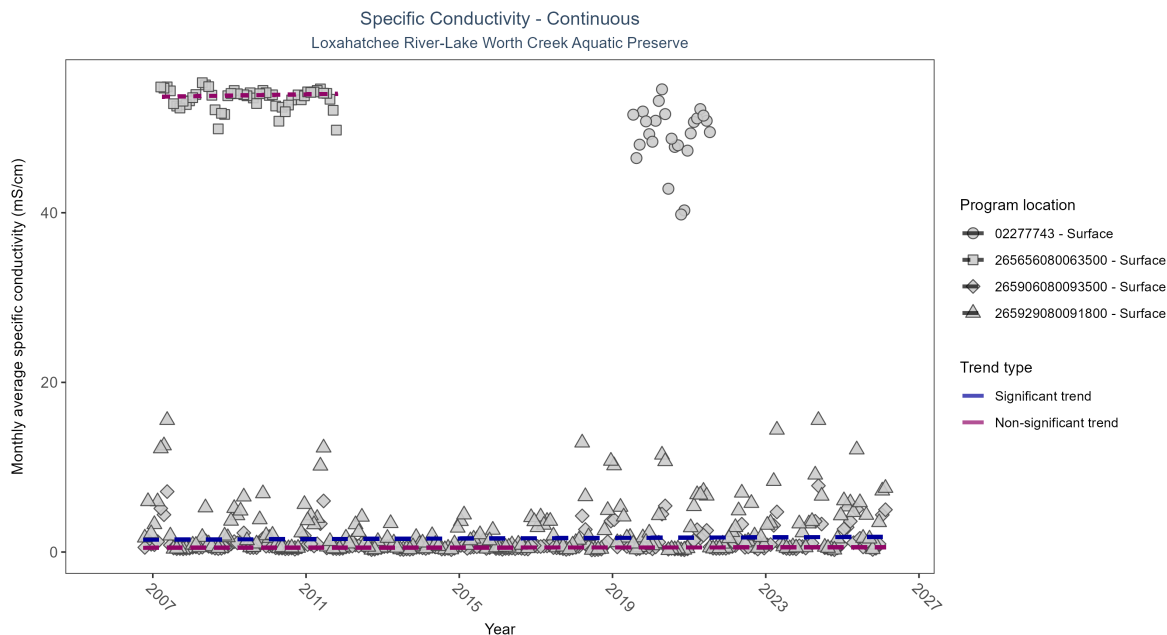


Table 10: Seasonal Kendall-Tau Results - Specific Conductivity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02277743	Insufficient data to calculate trend	1425	3	2019 - 2021	50.10	-	-	-	-
265656080063500	No significant trend	2550	5	2007 - 2011	53.90	0.04	53.66	0.08	0.9336
265906080093500	No significant trend	13578	21	2006 - 2026	0.56	0.07	0.5	0	0.1712
265929080091800	Significantly increasing trend	6830	21	2006 - 2026	0.82	0.11	1.43	0.02	0.0303

At one program location, monthly average specific conductivity increased by 0.02 ms/cm per year. No detectable change in monthly average specific conductivity was observed at two locations. There was insufficient data to fit a model for one location.

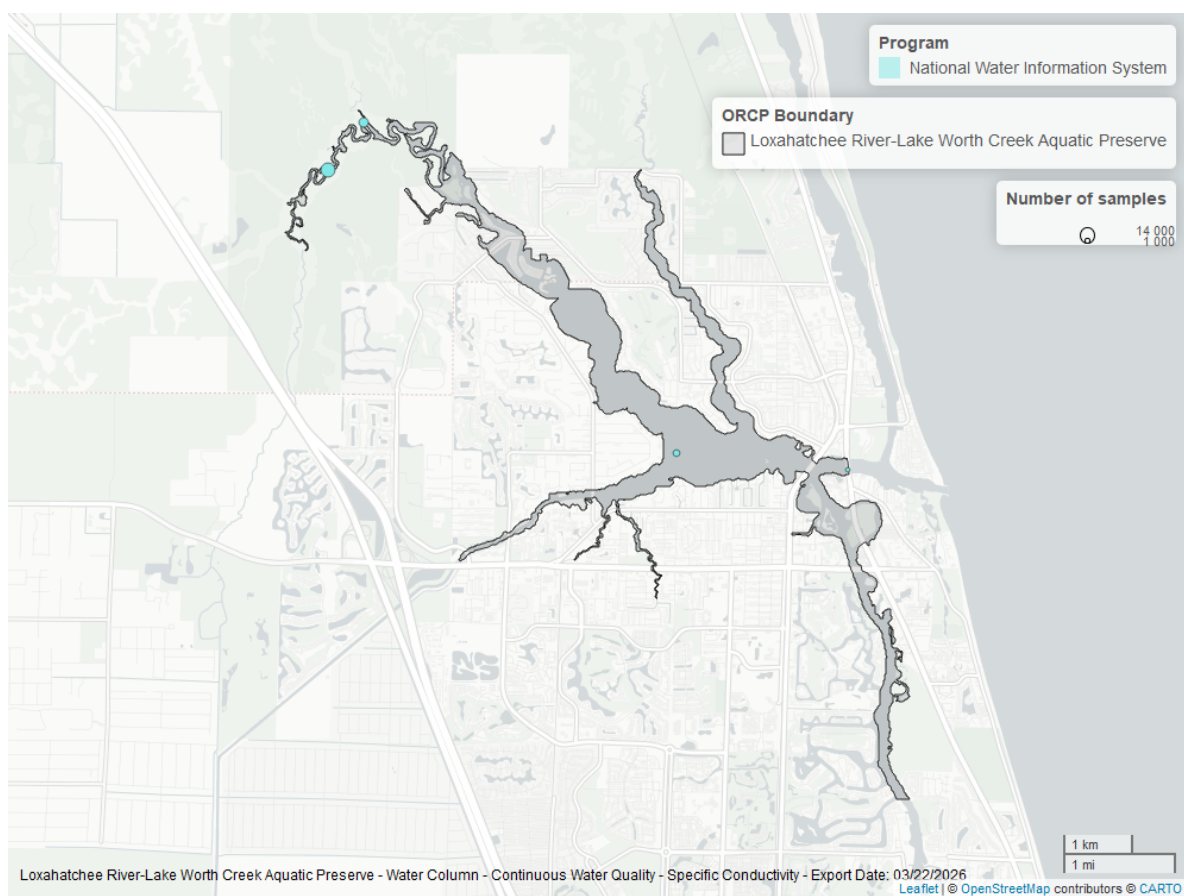


Figure 19: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Discrete

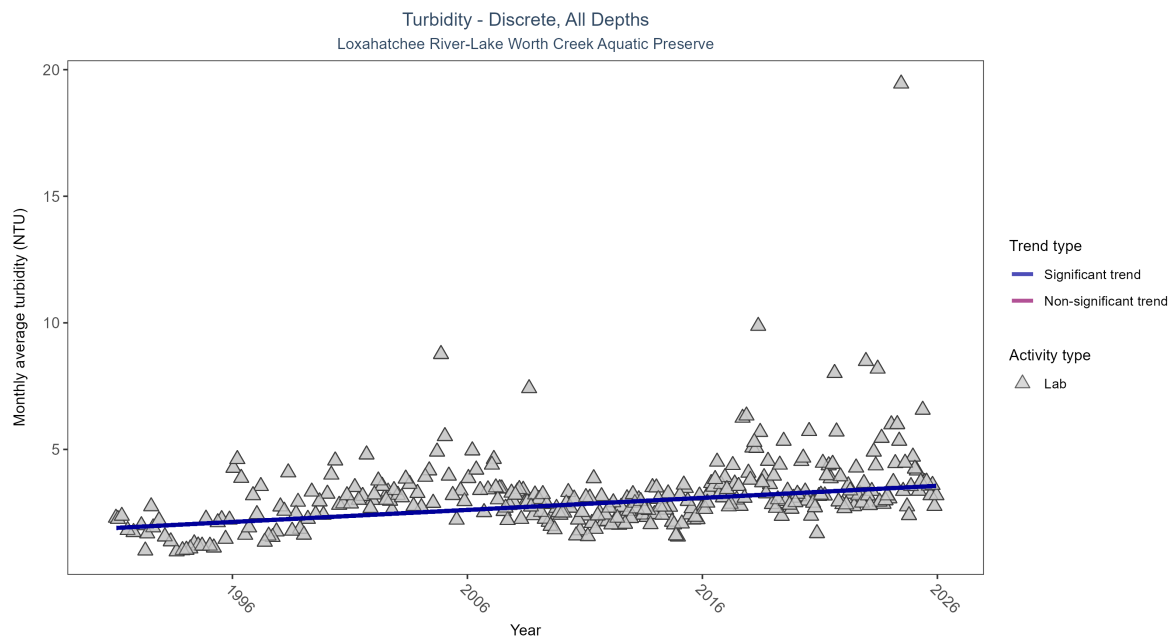


Figure 20: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 11: Seasonal Kendall-Tau Results for - Turbidity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	4111	35	1991 - 2025	2.7	0.3443	1.89695	0.04757	0

Monthly average turbidity increased by 0.05 NTU per year, indicating a decrease in water clarity.

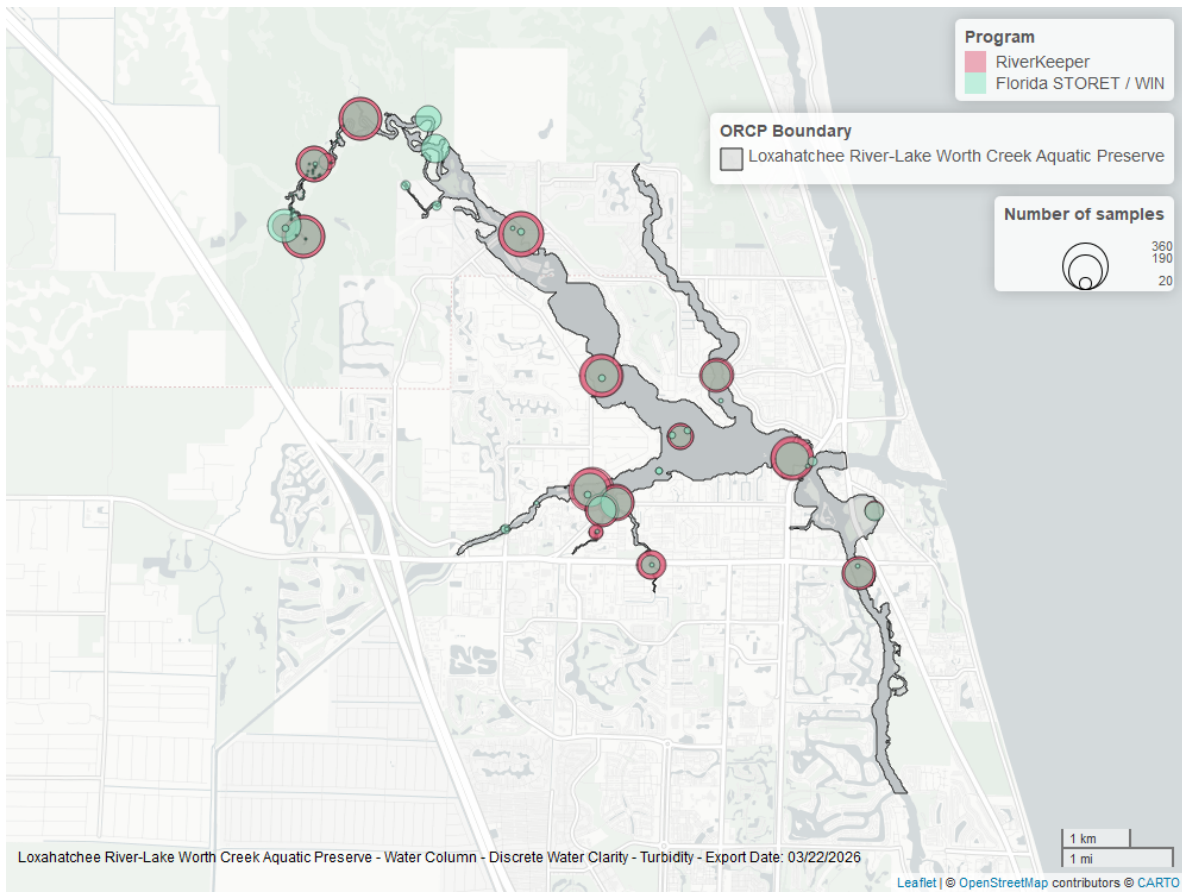


Figure 21: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

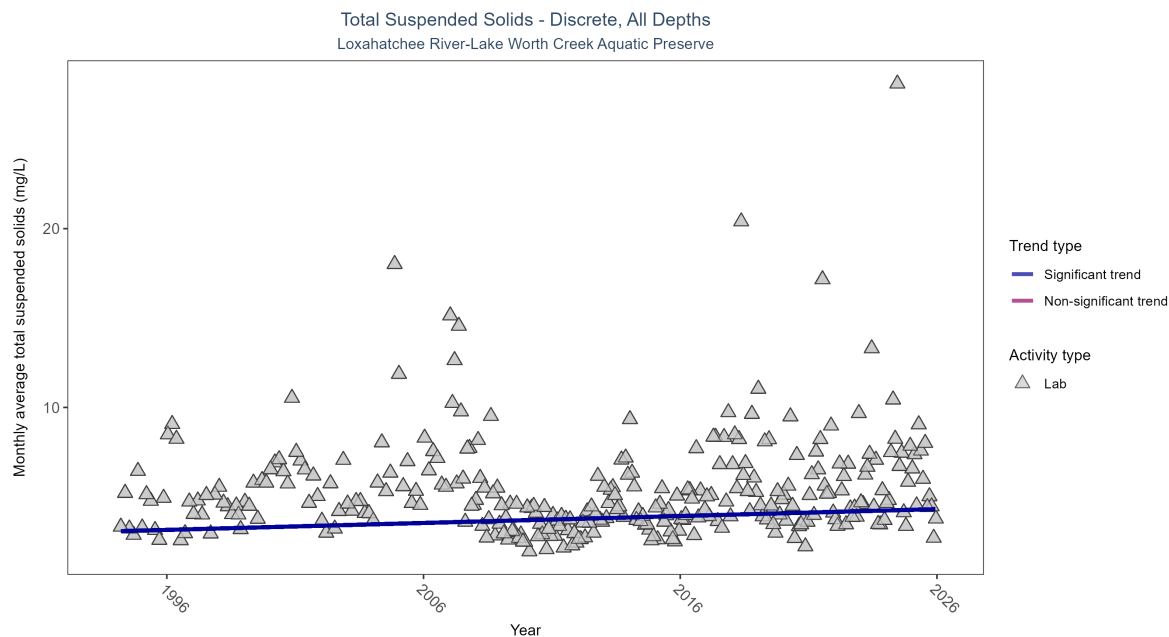


Figure 22: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 12: Seasonal Kendall-Tau Results for - Total Suspended Solids

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	3885	32	1994 - 2025	4	0.14318	3.07074	0.03874	0.0052

Monthly average total suspended solids increased by 0.04 mg/L per year, indicating a decrease in water clarity.

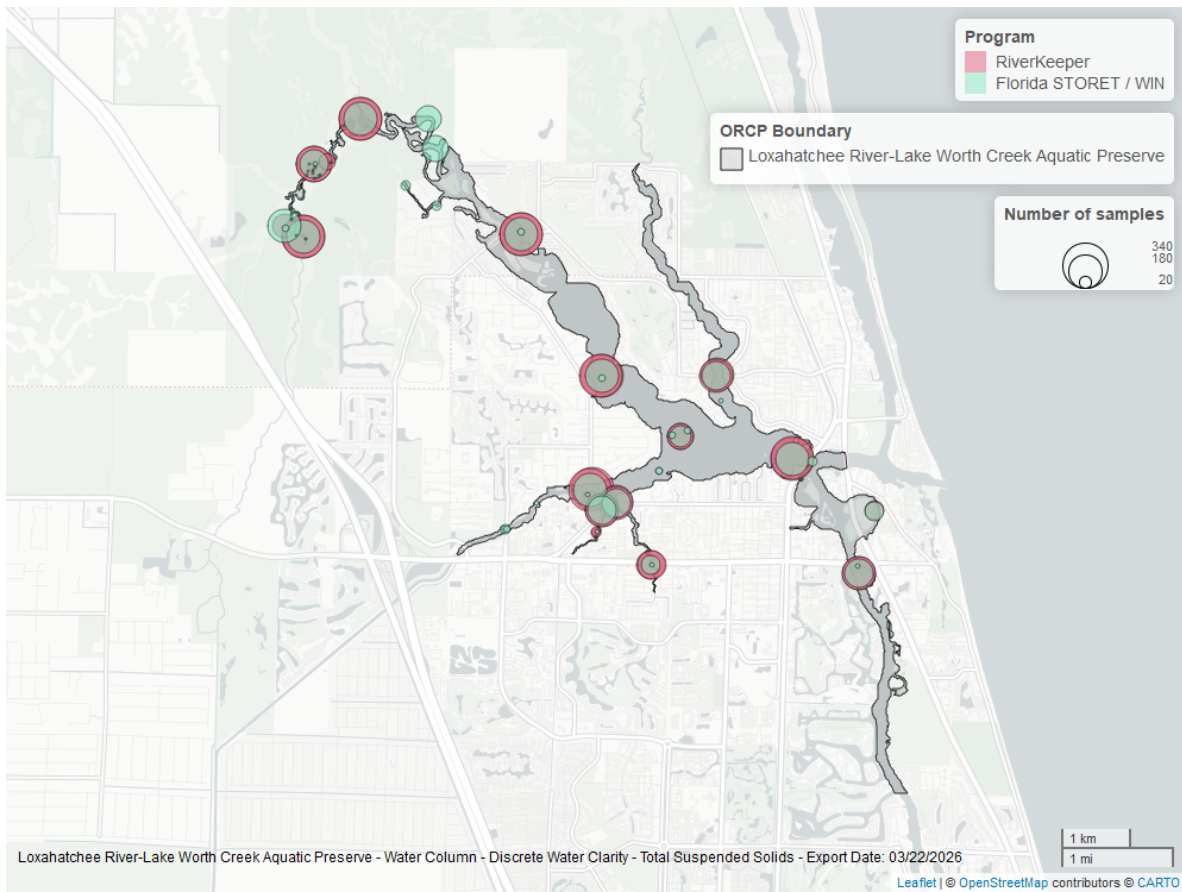


Figure 23: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

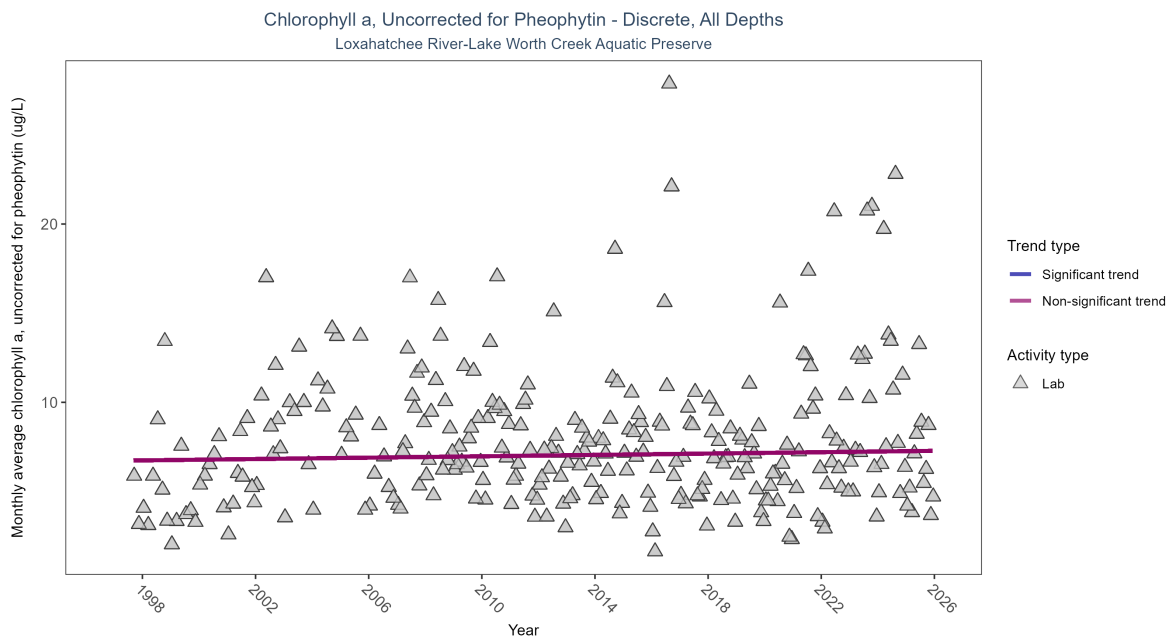


Figure 24: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 13: Seasonal Kendall-Tau Results for - Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	3826	29	1997 - 2025	5.7	0.04021	6.72414	0.01916	0.3162

Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 1997 and 2025.

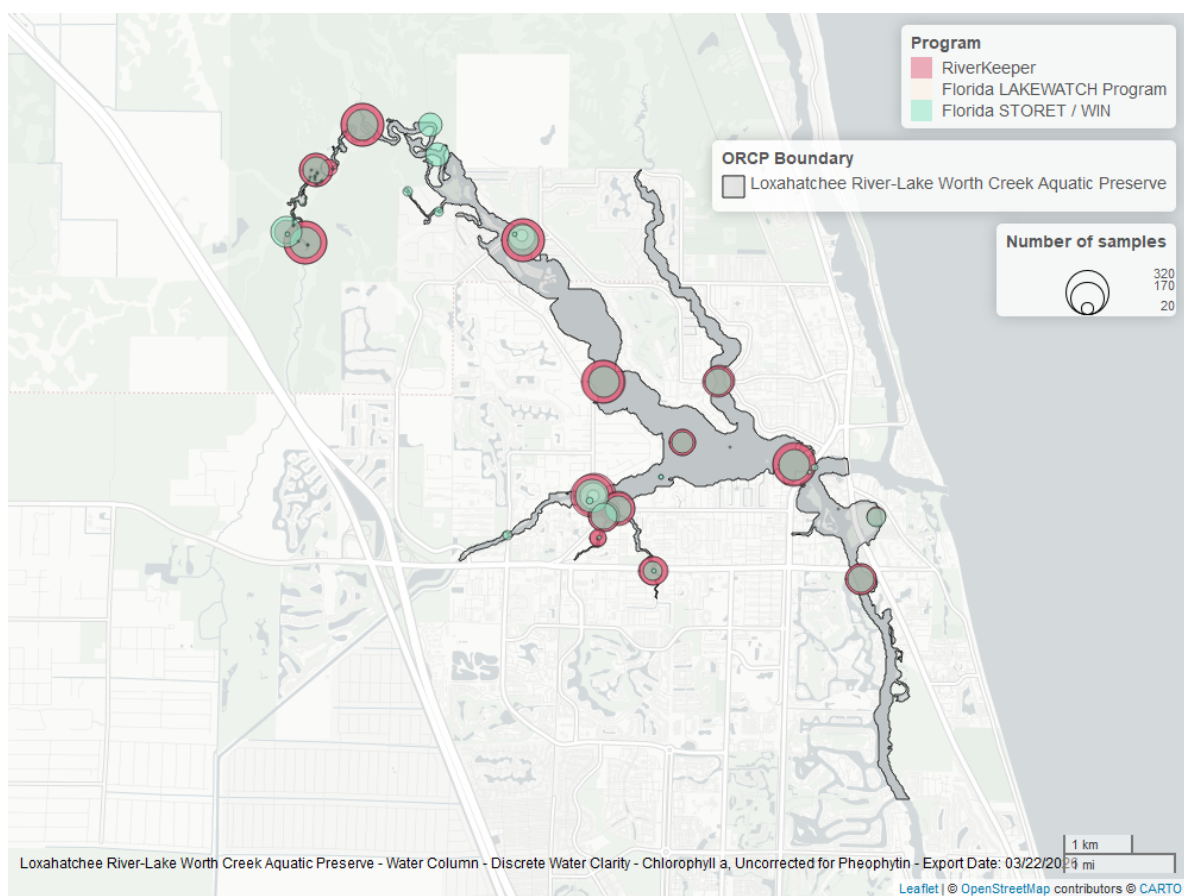


Figure 25: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

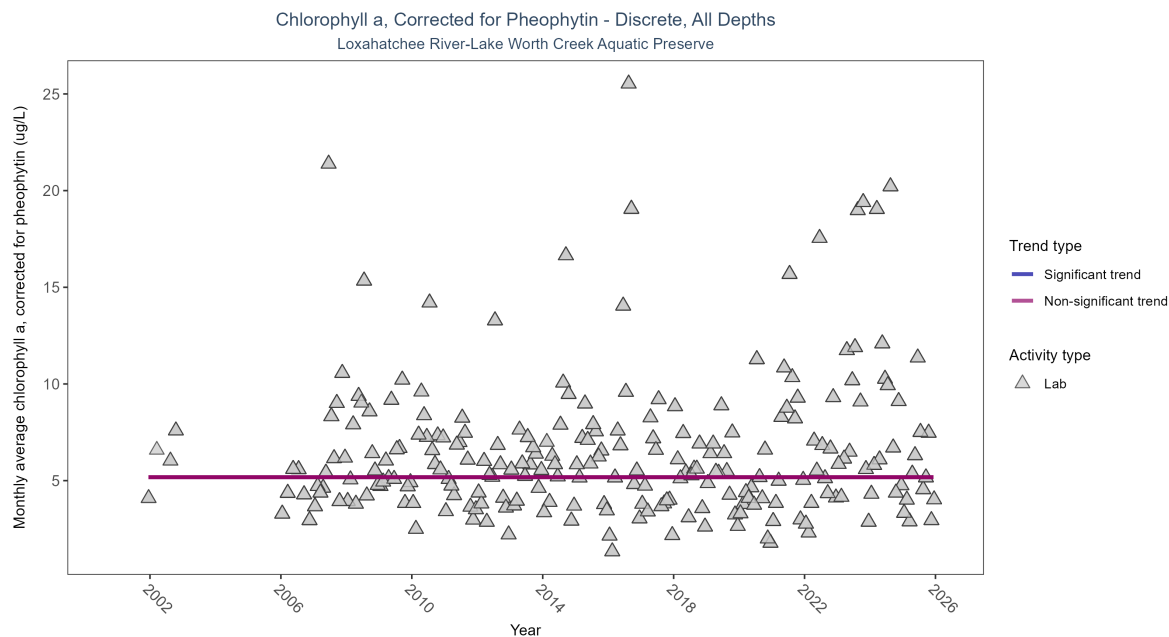


Figure 26: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 14: Seasonal Kendall-Tau Results for - Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	3014	22	2001 - 2025	4.5	0.00113	5.17854	-0.00003	0.9924

Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2001 and 2025.

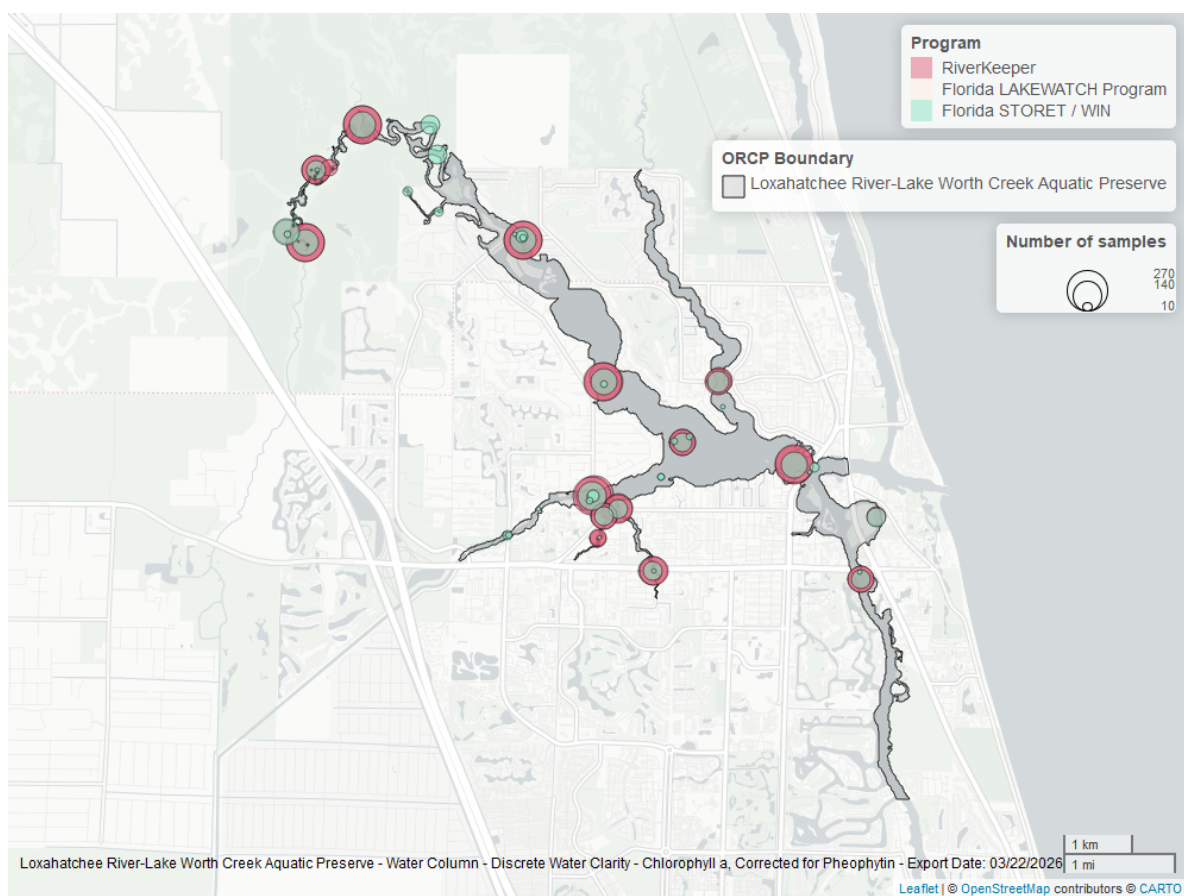


Figure 27: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

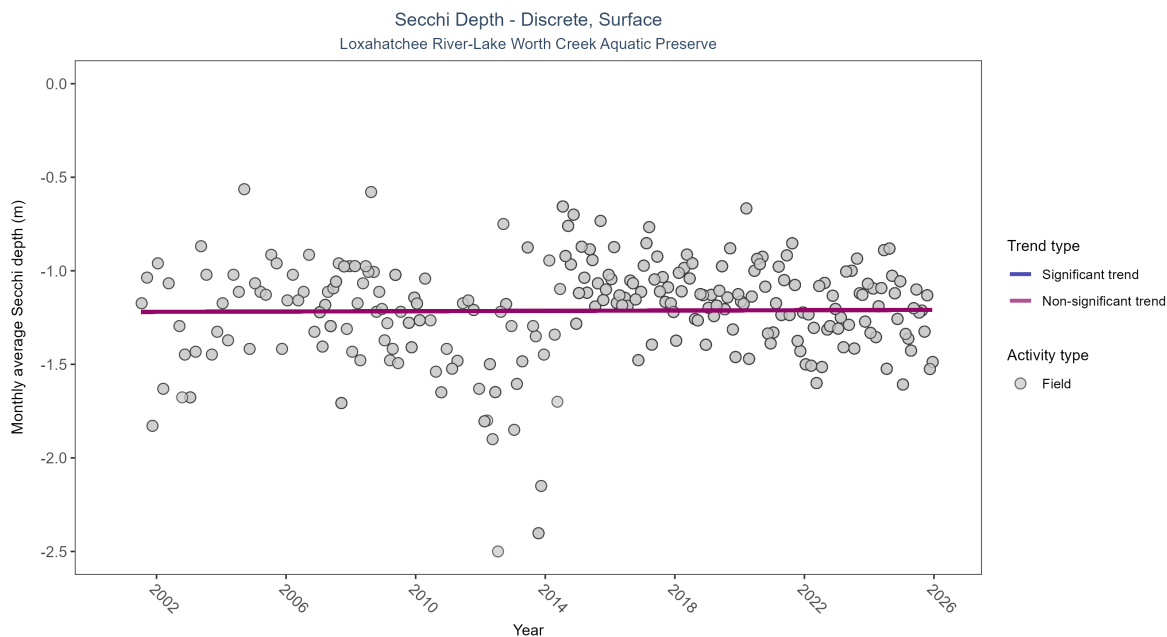


Figure 28: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 15: Seasonal Kendall-Tau Results for - Secchi Depth

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	2596	25	2001 - 2025	-1.1	0.01596	-1.21974	0.00043	0.8458

Secchi depth showed no detectable trend between 1997 and 2025.

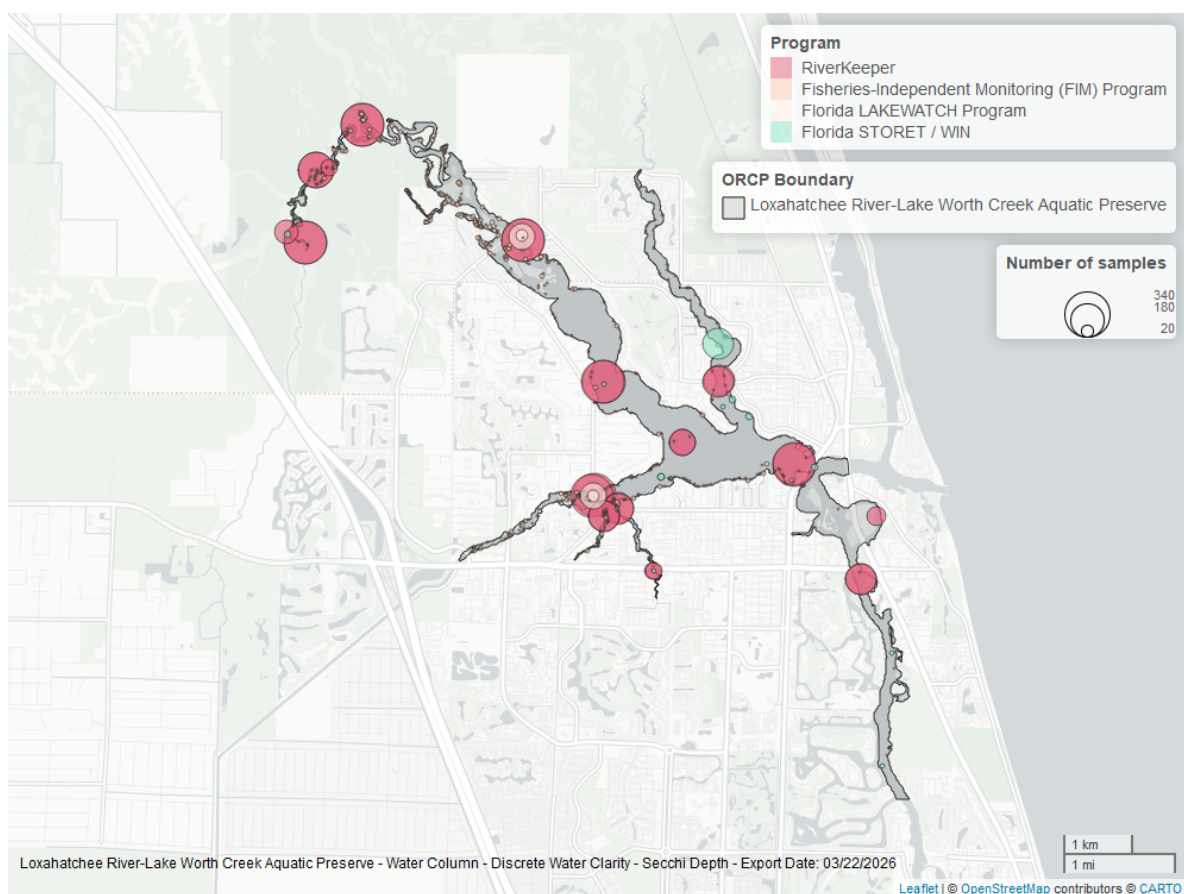


Figure 29: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

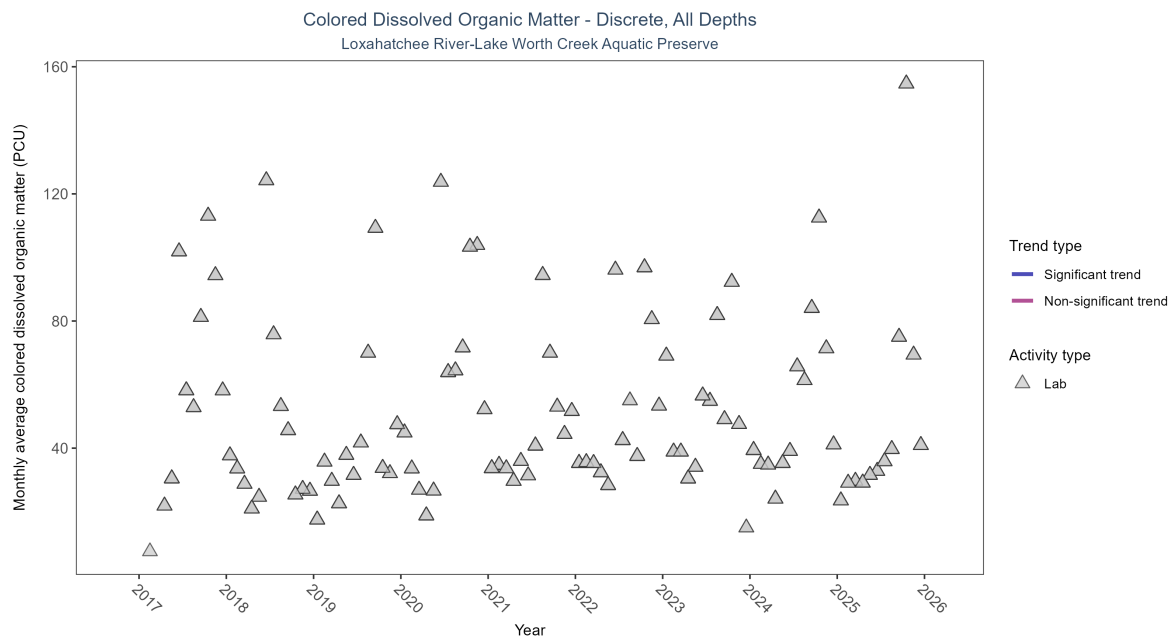


Figure 30: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 16: Seasonal Kendall-Tau Results for - Colored Dissolved Organic Matter

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data to calculate trend	1159	9	2017 - 2025	45	-	-	-	-

There was insufficient data to fit a model for colored dissolved organic matter.

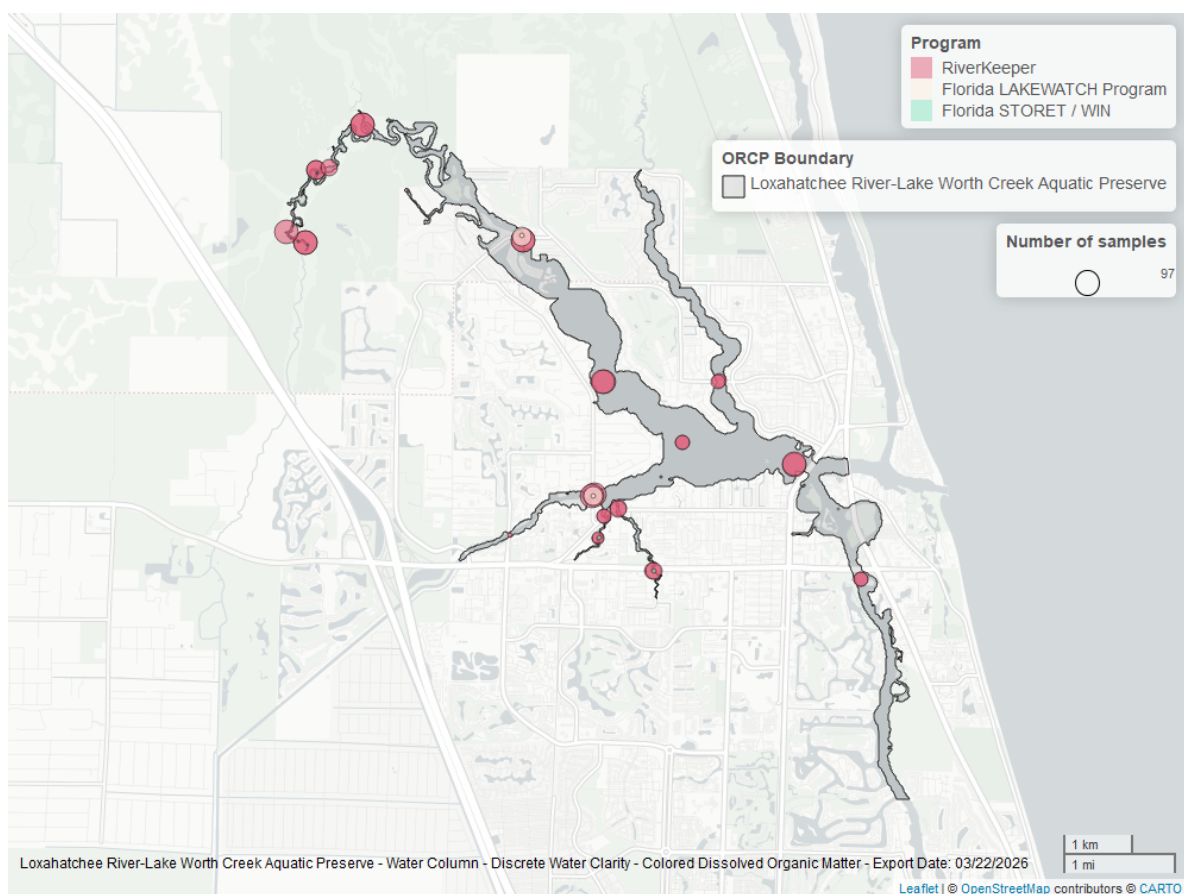


Figure 31: Map showing location of discrete water quality sampling locations within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.